

Data Science Job Roles

Module 10 Study Notes • Career Pathways

1. The MLDLC and Job Specialization

In the previous module, we discussed the Machine Learning Development Life Cycle (MLDLC), which encompasses everything from data gathering to deploying an API on a live server. Because this pipeline is so complex, a single person cannot handle it in a large enterprise. The pipeline is therefore broken down into four distinct job roles:

- **Data Engineer:** Handles Data Gathering and Storage.
- **Data Analyst:** Handles Exploratory Data Analysis (EDA) and Reporting.
- **Data Scientist:** Handles Feature Engineering and Model Training.
- **Machine Learning (ML) Engineer:** Handles Model Deployment and Operations (MLOps).

2. Data Engineer

The "Plumber" of the data ecosystem. Their job is to ensure clean, accessible data is available on the table for analysts and scientists to use.

- **Core Responsibility:** Converting running databases (OLTP - Online Transaction Processing) into analytical databases (OLAP - Online Analytical Processing, or Data Warehouses) using ETL (Extract, Transform, Load) pipelines. This ensures analytical queries don't crash the live website.
- **Key Skills:** Hardcore software engineering, Database Management (SQL/NoSQL), Distributed Systems, Cloud Platforms (AWS, GCP), and Big Data tools (Hadoop, Apache Spark).
- **Focus:** They do not care about the business domain (e.g., they don't need to know how cricket works to handle cricket data). They care purely about the technical flow of data.

3. Data Analyst (and Business Analyst)

The "Storyteller." Their job is to look at the *past* data and figure out what happened and why it happened.

- **Core Responsibility:** Cleaning data, running descriptive analytics, and creating visual dashboards to present to management (e.g., "Why did our sales drop last quarter?").
- **Key Skills:** SQL, Data Visualization tools (Tableau, PowerBI), strong statistical knowledge, and most importantly, Data Storytelling and Communication skills.

- **Focus:** They require very high "Business Acumen." If you are analyzing banking data, you must deeply understand the banking domain.

4. Data Scientist

The "Predictor." If a Data Analyst looks at the past, a Data Scientist looks into the *future*. A classic definition of a Data Scientist is: *"Someone who is better at statistics than any software engineer, and better at software engineering than any statistician."*

- **Core Responsibility:** Predictive modeling. They take the data, perform feature engineering, and train complex ML algorithms to predict future outcomes (e.g., "What products will sell best next month?").
- **Key Skills:** Advanced mathematics (Linear Algebra, Calculus), Probability & Statistics, Machine Learning Algorithms, Python/R, and strong business sense.
- **Focus:** They are the core intellectual force behind the ML model, bridging the gap between raw data analysis and predictive software.

5. Machine Learning (ML) Engineer

The "Deployer." Data Scientists often lack the software engineering skills to integrate their models into production, and traditional Software Engineers lack the knowledge of how ML models behave. The ML Engineer sits between them.

- **Core Responsibility:** Taking the trained model from the Data Scientist, wrapping it in an API, deploying it to a cloud server, and setting up automated pipelines for scaling and retraining (MLOps).
- **Key Skills:** Cloud Architecture, System Design, Containerization (Docker/Kubernetes), and strong software engineering (Python, C++).
- **Focus:** High technical execution with a focus on system reliability, with less emphasis on business storytelling.

6. Summary Comparison

Role	Primary Focus	Key Skills	Business Domain Knowledge Needed?
------	---------------	------------	-----------------------------------

Data Engineer	Data Infrastructure & Pipelines	Big Data, SQL, Cloud, System Design	Low
Data Analyst	Past Insights & Dashboards	SQL, Tableau, Statistics, Storytelling	Very High
Data Scientist	Future Predictions & ML Models	Math, ML Algorithms, Python	High
ML Engineer	Model Deployment & MLOps	Cloud, API, Docker, Software Engineering	Medium